

When Does Self-Training Collapse on Protein Language Model Embeddings?

A Controlled Study of Class Imbalance, Confidence Calibration, and
Masking-Based Consistency Regularization

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Abstract

Self-training (pseudo-labeling) is the simplest and most widely used semi-supervised learning (SSL) method, but when and why it collapses—performance falling far *below* a supervised-only baseline, not just short of it—remains poorly characterized quantitatively. We study this using protein language model (PLM) embeddings as a controlled testbed, directly manipulating class imbalance on frozen ESM-2 embeddings (DeepLoc 2.0 subcellular localization) to isolate its causal role. Across a fine-grained sweep and multiple seeds per condition, collapse concentrates in a reproducible “danger zone” (roughly 8–20% positive rate), confirmed to reflect training-pool imbalance itself rather than train-test distribution mismatch. Applying seed-level statistical rigor to calibration, we show Expected Calibration Error (ECE) reliably rises whenever self-training collapses ($r = -0.91$, partial $r = -0.92$ controlling for imbalance level), while pre-training ECE does not predict later collapse—indicating miscalibration is a necessary but not sufficient precursor. Comparing remedies, calibrated self-training and confidence-threshold tuning both give only partial relief, while explicitly balancing which pseudo-labels are accepted across classes eliminates collapse entirely and slightly exceeds the supervised baseline; masking-based consistency regularization, which never thresholds into hard pseudo-labels, is robust throughout without any calibration correction. We further show the danger zone reproduces at a four-times-larger PLM scale, is not explained by embedding geometry, and persists against binary-adapted DARP and CReST baselines from the class-imbalanced SSL literature—both underperform our class-balanced selection, with DARP collapsing even more than naive self-training despite an oracle advantage unavailable in practice. We frame protein PLMs as a case study for a general SSL failure mode, and release our controlled-imbalance protocol, failure-regime criteria, and remedy comparisons for use in other domains.

1 Introduction

Self-training (pseudo-labeling) [Lee, 2013] is the simplest and most widely deployed form of semi-supervised learning (SSL): a model trained on available labels labels its own confident predictions on unlabeled data, which are then folded back into training. It is well known that this procedure is vulnerable to *confirmation bias*—early mistakes reinforcing themselves over successive rounds—but the literature’s characterization of this vulnerability is largely qualitative. *When*, precisely, does self-training merely underperform a better SSL method, and when does it actively collapse below a supervised-only baseline that ignores the unlabeled data entirely? This question matters practically: a practitioner deciding whether to add self-training to a pipeline needs to know not just that failure is possible, but under what conditions it becomes likely.

We study this question using protein language models (PLMs) as a controlled testbed. PLMs such as ESM-2 [Lin et al., 2023], pretrained via masked language modeling on hundreds of millions of

unlabeled sequences, produce embeddings that transfer well to many downstream tasks, while labeled data for a specific property of interest—such as experimentally verified subcellular localization—is often scarce. This combination of abundant unlabeled structure and scarce labels is exactly the setting SSL targets [Van Engelen and Hoos, 2020], and multi-label subcellular localization prediction [Thummuluri et al., 2022] additionally exhibits severe, naturally occurring class imbalance across its ten compartment categories. We first show, empirically, that naive self-training on frozen ESM-2 embeddings collapses catastrophically under label scarcity in this setting (Section 4). We then go beyond this single observation and ask what, specifically, is responsible: is it the class imbalance, the label scarcity, some interaction of both, or something particular to this dataset?

From observation to controlled experiment. A single dataset comparison cannot answer this question, because class imbalance, label scarcity, class count, and task difficulty are all entangled. We therefore design a *controlled* experiment: we fix the embedding space, the classifier architecture, and the evaluation set, and directly manipulate class imbalance by subsampling positive and negative examples for one localization category (Nucleus) to hit a chosen target positive rate, independent of the natural label distribution. Sweeping this manipulated imbalance level against several label fractions, and repeating every condition across multiple random seeds, lets us isolate imbalance’s causal contribution and—critically—quantify how *consistently* collapse occurs at each level, rather than reporting a single run that may or may not generalize.

From qualitative narrative to quantified mechanism. We trace collapse to *confidence miscalibration*: when the classifier’s predicted probabilities do not reflect true correctness likelihoods, a fixed confidence threshold for pseudo-label acceptance admits large numbers of incorrect, overconfident pseudo-labels, which self-reinforce over self-training rounds. We measure this directly with Expected Calibration Error (ECE), tracked round-by-round on a held-out test set, and—rather than relying on visually correlated trends—we apply an explicit, seed-level statistical test of the causal chain: does rising ECE actually co-occur with collapse, run by run, and how often?

From single remedy to comparative diagnosis. A calibration-based causal story is only as convincing as the interventions it predicts. We therefore go beyond diagnosis and directly compare three candidate remedies for the identified failure mode—temperature-scaled calibration, confidence-threshold tuning, and class-balanced pseudo-label selection—asking which one, if any, actually prevents collapse, and separately verify with a matched-test-distribution control that the identified danger zone is not an artifact of train-test distribution mismatch.

Contributions.

- We provide a controlled empirical demonstration, with an explicit, reproducible class-imbalance manipulation protocol, that self-training on PLM embeddings collapses in a specific, identifiable, non-monotonic regime of class imbalance and label scarcity—not simply “the more imbalanced, the worse”—and rule out, via a matched-test-distribution control, that this danger zone is an artifact of train-test prior shift rather than training-pool imbalance itself.
- We introduce a quantitative, threshold-based classification of self-training outcomes (baseline-degenerate / self-training collapse / stable / partial degradation) and apply it at the level of individual random seeds, showing that averaged results alone can both overstate and understate how reliably collapse occurs.
- We quantify, rather than merely illustrate, the calibration-collapse causal chain: across 50 controlled runs, ECE escalation during self-training correlates strongly with the performance gap ($r = -0.91$), a relationship that remains equally strong after statistically controlling for imbalance level (partial $r = -0.92$)—ruling out imbalance level as a common-cause confound—while ECE measured before self-training begins does not predict later collapse (AUC= 0.55),

indicating the relationship concerns the *dynamics* of calibration during self-training rather than a static property of the initial model.

- We directly compare three candidate remedies for the identified failure mode—calibrated self-training, confidence-threshold tuning, and class-balanced pseudo-label selection—and find that while calibration meaningfully reduces collapse frequency (75%→38% of seeds), explicitly balancing which pseudo-labels are accepted across classes is markedly more effective, eliminating collapse entirely and slightly exceeding the supervised baseline, indicating that biased pseudo-label selection under imbalance, not miscalibration alone, is the more direct lever.
- We show that masking-based consistency regularization, which reuses a PLM’s own pretraining mechanism as a label-free augmentation rather than relying on hard pseudo-labels, remains robust throughout the identified danger zone without any explicit calibration correction, offering a simple, PLM-native alternative to naive self-training.
- We test whether our findings generalize beyond a single small PLM checkpoint by repeating the fine-grained imbalance sweep, representative-level verification, and ECE-F1 causal-chain analysis with a second, four-times-larger ESM-2 checkpoint (35M parameters) under an identical protein sample and data split: the danger zone, its approximate location, and the calibration-collapse relationship all reproduce, with collapse severity mildly attenuated but not eliminated at the larger scale.
- We test and rule out an alternative, purely geometric explanation for the danger zone: four embedding-space diagnostics (pool and within-class anisotropy, class-centroid cosine similarity, and effective rank) show no relationship with collapse severity that survives controlling for imbalance level and correcting for testing multiple metrics simultaneously, indicating the danger zone reflects imbalance-driven pseudo-labeling dynamics rather than a property of the embedding geometry itself.
- We directly compare class-balanced pseudo-label selection against binary-classification adaptations of two established class-imbalanced SSL methods, DARP and CReST, and find both underperform it in this setting—including a DARP-style baseline given an oracle advantage unavailable in practice—indicating that per-round, per-class confidence ranking, rather than aggregate distribution alignment or a progressive curriculum, is the more effective lever on this failure mode.

2 Related Work

Semi-supervised subcellular localization prediction. Prior work has explored SSL for multi-label protein subcellular localization using predictive clustering trees [Alcantara et al., 2025], showing that unlabeled data can help in this task. Our work asks a different, complementary question: not whether SSL can help on this task, but under what precise, controllable conditions a widely used SSL method (self-training) instead fails catastrophically, and why.

Class imbalance in subcellular localization. The severe class imbalance across the ten DeepLoc compartments has been addressed directly at the data level via multi-label oversampling on image-derived features [Rana et al., 2023]. Our findings are complementary and mechanistic: rather than proposing a new imbalance-correction method, we directly manipulate imbalance as an

experimental variable to establish its causal role in *specifically corrupting the pseudo-labeling step* of self-training, and characterize the resulting failure region quantitatively.

Class-imbalanced semi-supervised learning. Beyond subcellular localization specifically, a substantial general ML literature addresses class imbalance in SSL directly, including distribution-aligning pseudo-label refinement (DARP) [Kim et al., 2020] and class-rebalancing self-training that explicitly re-weights or re-samples which pseudo-labels are accepted per class (CReST) [Wei et al., 2021]. Beyond echoing this literature’s insight conceptually, we directly implement binary-classification adaptations of both methods and compare them against our class-balanced pseudo-label selection at the identified danger zone (Section 4.9): both underperform our simpler method in this setting, including a DARP-style baseline given an oracle advantage (knowledge of the true class ratio) unavailable in realistic deployments, where the target marginal must itself be estimated. We report this comparison together with its important caveats—the binary-classification simplification of both methods, and DARP’s oracle assumption—rather than as a claim that our method dominates these approaches in general. Our contribution remains complementary to this literature: to mechanistically explain *why* naive self-training fails under imbalance via a quantified calibration pathway, and to test where a range of remedies, from our own to established literature baselines, sit relative to that mechanism, rather than to introduce or claim general superiority as a new class-imbalanced SSL method.

PLM embeddings for localization. ESM-2 embeddings have been used for subcellular localization in fully supervised multi-task settings, e.g. jointly predicting Gene Ontology annotations and localization [Bai et al., 2024]. Our setting differs in that we study the label-scarce regime explicitly and treat unlabeled sequences as a first-class resource via SSL, rather than using auxiliary fully labeled tasks.

Masking-based latent consistency. The idea of enforcing consistency between representations of masked and unmasked inputs has recently appeared in PLM *pretraining* [Ofer et al., 2026], where it is used purely self-supervised on unlabeled sequence corpora with no downstream task. We repurpose the same underlying idea—exploiting a PLM’s own masking mechanism—as a *downstream*, label-scarce consistency-regularization signal, and show it remains robust precisely in the regime where self-training collapses.

Confidence calibration. It is well established that modern neural networks are frequently overconfident [Guo et al., 2017], and that this can be corrected cheaply via temperature scaling without retraining the underlying model. Calibration has been studied extensively for supervised prediction; our contribution is to quantify its role as a *necessary precursor* to self-training collapse under class imbalance, using an explicit, seed-level conditional-probability test rather than a qualitative or purely visual correlation—and to directly test, rather than assume, how much of the failure calibration alone actually explains (Section 4.8).

3 Method

3.1 Protein Embeddings

We encode each protein sequence with a pretrained ESM-2 model [Lin et al., 2023] (`facebook/esm2_t6_8M_UR50D`, 8M parameters), taking the mean of the final-layer token representations over non-padding positions as a fixed-length embedding. The PLM itself is kept frozen throughout; we do not fine-tune its weights. Using a small, fast checkpoint lets us run the extensive multi-seed controlled experiments in Section 4. To test whether our findings are an artifact of this particular small checkpoint, Section 3.10 repeats the controlled single-category experiments with a second, larger checkpoint under an identical data split; we treat validating the mechanism at

substantially larger PLM scale (e.g., hundreds of millions of parameters) as further future work (Section 5).

3.2 Problem Setup

We use DeepLoc 2.0 [Thummuluri et al., 2022], which labels each protein with a subset of 10 subcellular compartments (multi-label). We work with two related experimental tracks built on the same underlying data and embeddings.

Multi-label track. We sample $N = 1500$ sequences from the DeepLoc 2.0 training split (`random_state=42` for reproducibility), holding out a fixed 20% (300 sequences) as the test set and using the remaining 1200 as an unlabeled/labeled pool. To simulate label scarcity, we retain only a fraction $f \in \{10\%, 25\%, 50\%, 100\%\}$ of this pool’s labels and treat the remainder as unlabeled. We report micro-F1, macro-F1, and Hamming loss.

Controlled single-category track. To isolate the causal role of class imbalance (Section 3.4), we binarize one compartment, Nucleus, as the prediction target. We chose Nucleus because its natural positive rate ($\approx 40\%$) is high enough to support subsampling down to very low or back up to near-balanced positive rates without running out of positive examples. This track reuses the same fixed 300-example test split as the multi-label track (restricted to the Nucleus label), so results on the two tracks are directly comparable.

3.3 Self-Training Baseline

We extend standard confidence-thresholded self-training [Lee, 2013] to the multi-label setting by treating each (protein, compartment) pair as an independently maskable cell: a cell is accepted as a pseudo-label once the classifier’s predicted probability exceeds a confidence threshold $\tau = 0.9$ in either direction, and training proceeds via a masked binary cross-entropy loss over all currently-known cells (originally labeled or pseudo-labeled), repeated for up to 5 rounds (fewer if no new confident pseudo-labels are found). In the controlled single-category track, the same procedure is applied to the single Nucleus target. All classifiers are two-layer MLPs (64 hidden units, ReLU) trained with Adam (learning rate 10^{-3} , 150 epochs) on top of the frozen ESM-2 embedding.

3.4 Controlled Class-Imbalance Manipulation

A single comparison across naturally occurring tasks (e.g., a roughly balanced binary task versus a severely imbalanced multi-label one) confounds class imbalance with other differences (class count, task difficulty, label semantics). To isolate imbalance as a variable, we construct an artificially imbalanced training pool of fixed size 800 for the Nucleus target: given a target positive rate r , we subsample positive and negative examples from the (unchanged) embedding pool without replacement to hit r as closely as the available examples allow. In our main experiments, *the held-out test set is never resampled*—it always reflects the natural label distribution—so that all reported metrics measure generalization to the real data distribution under a training pool whose imbalance we control; Section 4.5 additionally reports a matched-test-distribution control condition used to isolate training-pool imbalance from train-test distribution mismatch. We sweep r both coarsely (5 levels, paired with 6 label fractions, single seed, as an initial pilot) and, as our primary evidence, on a fine-grained grid (10 levels between 6% and 30%, at a fixed label fraction of 10%, repeated over 5 random seeds per level, plus 8-seed replication at three representative levels) to obtain seed-level statistics rather than single-run point estimates.

3.5 Diagnosing Failure via Calibration

We measure calibration using Expected Calibration Error (ECE), which bins predictions by confidence $\max(p, 1 - p)$ and compares average confidence to empirical accuracy within each bin:

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{conf}(b) - \text{acc}(b)|.$$

In the controlled track, we compute ECE on the fixed held-out test set *before* each round’s pseudo-labeling step, giving a round-by-round calibration trajectory that we can align against the corresponding test-F1 trajectory for the same run.

3.6 Quantitative Failure-Regime Classification

To avoid characterizing results by unprincipled, post-hoc description (e.g., informally calling one condition “safe” and another “dangerous”), we define four regimes from two directly measured quantities, the mean baseline F1 and mean self-training F1 at a given condition, using fixed, pre-specified thresholds:

- **A (baseline-degenerate):** $\text{F1}_{\text{base}} < 0.10$. The supervised baseline itself fails to learn a usable signal; comparing self-training against it is not meaningful.
- **B (self-training collapse):** $\text{F1}_{\text{base}} \geq 0.10$ and $\text{F1}_{\text{self}} \leq 0.5 \times \text{F1}_{\text{base}}$. The baseline learns a usable signal, but self-training destroys more than half of it.
- **C (stable):** $\text{F1}_{\text{base}} \geq 0.10$ and $\text{F1}_{\text{self}} \in [0.8, 1.2] \times \text{F1}_{\text{base}}$. Self-training is statistically indistinguishable from the baseline.
- **D (partial degradation):** conditions satisfying none of the above—self-training is measurably worse than baseline, but not by the factor required for regime B.

We apply this classification both to condition-level means and, separately, to each individual random seed, and report the *fraction of seeds* independently classified as regime B at each imbalance level—since a mean can mask substantial seed-to-seed variability. We similarly define an ECE-escalation criterion, $\text{ECE}_{\text{last round}} - \text{ECE}_{\text{round 1}} \geq 0.05$ (an absolute rather than relative threshold, since relative thresholds are unstable when the starting ECE is near zero), and use it in Section 4.7 to test the calibration-collapse causal chain at the seed level, including a partial-correlation reanalysis that controls for imbalance level and a test of whether pre-training (round-1) ECE alone predicts later collapse. All thresholds above $(0.10, 0.5 \times, [0.8, 1.2] \times, 0.05)$ are pre-specified design choices rather than fit to the data; we report sensitivity to these choices as a direction for future work (Section 5).

3.7 Calibrated Self-Training

As a principled fix motivated by the diagnosis above, we implement *calibrated self-training*: at each self-training round, we hold out a stratified 25% subset of the currently labeled data (sampled separately from positive and negative examples, to preserve minority-class representation) purely for calibration, fit a single scalar temperature T by grid search over $T \in [0.1, 5.0]$ to minimize the negative log-likelihood of $\sigma(z/T)$ against the held-out labels, and use $\sigma(z/T)$ in place of raw $\sigma(z)$ when deciding pseudo-label acceptance. Because temperature scaling is a monotonic transform, it changes which pseudo-labels are confident enough to accept, but not the hard label (> 0.5) a given example would receive. Withholding data for calibration means this method trains its classifier on

fewer labeled examples than the other compared methods at the same nominal label budget, a cost we make explicit in Section 5.

3.8 Alternative Remedies: Threshold Tuning and Class-Balanced Pseudo-Label Selection

Calibration is one candidate explanation for collapse, but a simpler alternative is that collapse is merely an artifact of the fixed confidence threshold $\tau = 0.9$ used throughout, or of not accounting for class imbalance when deciding *which* pseudo-labels to accept. We test both alternatives directly. First, we re-run naive (uncalibrated) self-training with $\tau \in \{0.7, 0.8, 0.9, 0.95, 0.99\}$, holding everything else fixed. Second, we implement *class-balanced pseudo-label selection*: rather than accepting every unlabeled example whose confidence exceeds τ in either direction, we separately rank candidate positive and negative pseudo-labels by confidence and accept the top- k from each side, where k is the size of the smaller of the two candidate pools—enforcing an equal number of newly accepted positive and negative pseudo-labels per round regardless of how the raw confident set is skewed. This is conceptually related to class-rebalancing ideas in the general imbalanced-SSL literature [Kim et al., 2020, Wei et al., 2021] (Section 2), applied here as a diagnostic for our failure mode rather than as a proposed new method.

3.9 Masking-Based Consistency Regularization

For each protein we compute two embeddings: the original ESM-2 embedding, and one computed after masking a random 15% of input tokens (at the tokenized-ID level) before the forward pass—directly reusing ESM-2’s own masked-language-modeling mechanism. We add a consistency loss that penalizes disagreement between the classifier’s predictions on the two embeddings for unlabeled proteins, with the original-embedding prediction treated as a stop-gradient target (following the Mean Teacher / FixMatch family of methods [Tarvainen and Valpola, 2017, Sohn et al., 2020]):

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda \cdot \text{MSE}(\sigma(z_{\text{masked}}), \text{sg}[\sigma(z_{\text{orig}})]).$$

Unlike self-training, this method never converts soft predictions into hard pseudo-labels, so it has no confidence threshold to be corrupted by miscalibration.

3.10 Scale Robustness: A Second, Larger PLM Checkpoint

To test whether our findings are specific to the small 8M-parameter checkpoint used elsewhere in this paper, we repeat the controlled single-category track—the fine-grained imbalance sweep, the representative-level 8-seed verification, and the ECE-F1 causal-chain analysis—using a second, four-times-larger ESM-2 checkpoint, `facebook/esm2_t12_35M_UR50D` (35M parameters). We reuse the identical `random_state=42` protein sample and train/pool/test split as the 8M experiments, changing only the embedding source, so that any difference between the two conditions is attributable to PLM scale rather than to a different random sample of proteins or a different data split.

3.11 Embedding Geometry Diagnostics

As an independent test of whether the danger zone is better explained by a property of the embedding space itself rather than by imbalance-driven pseudo-labeling dynamics, we compute four geometric diagnostics on the (unchanged) ESM-2 embeddings at each imbalance level and seed of the fine-grained sweep (Section 3.4), without any additional training: (1) *pool anisotropy*, the mean

pairwise cosine similarity across all embeddings in the imbalanced training pool; (2) *class anisotropy*, the same quantity computed separately within the positive and negative subsets; (3) *centroid cosine similarity*, the cosine similarity between the positive- and negative-class mean embedding vectors; and (4) *effective rank*, the participation ratio $(\sum_i s_i)^2 / \sum_i s_i^2$ of the singular values s_i of the pool’s embedding matrix, a continuous measure of how many dimensions carry non-trivial variance. High anisotropy and low effective rank have been associated with degraded discriminability of contextualized representations in NLP settings [Ethayarajh, 2019]; here we test whether they also predict where self-training collapses, using the same partial-correlation and predictive-power tests applied to the calibration causal chain (Section 4.7).

3.12 Binary-Adapted Class-Imbalanced SSL Baselines: DARP-style and CReST-style

To situate class-balanced pseudo-label selection (Section 3.8) relative to established class-imbalanced SSL methods directly, rather than only conceptually (Section 2), we implement binary-classification adaptations of two representative approaches. Both DARP [Kim et al., 2020] and CReST [Wei et al., 2021] were designed for multi-class, long-tailed classification and require simplification for our binary setting; we describe both adaptations and their limitations explicitly rather than claiming a faithful reproduction of the original methods.

DARP-style (distribution-aligning pseudo-label refinement). At each self-training round, we search over a scalar logit bias b (grid search over $[-6, 6]$ in steps of 0.05) and select the value minimizing $\left| \frac{1}{|U|} \sum_{i \in U} \sigma(z_i + b) - r_{\text{target}} \right|$, where U is the currently-unlabeled set; pseudo-label acceptance and the hard label are then both based on the bias-adjusted probability $\sigma(z_i + b)$ rather than the raw $\sigma(z_i)$. We set r_{target} to the *true* engineered positive rate r of the training pool (Section 3.4), which is known to us because we constructed the pool—an oracle simplification relative to real deployments, where the target marginal must itself be estimated from limited labeled data and is not necessarily known exactly. This gives the DARP-style baseline a built-in advantage over every other method we compare, none of which are told r .

CReST-style (progressive class-rebalancing curriculum). At round k of n_{rounds} , we compute a curriculum weight $\alpha_k = k / \max(1, n_{\text{rounds}} - 1)$ (so $\alpha_1 = 0$) and cap the number of majority-class pseudo-labels accepted that round at an interpolation between accepting all confident majority-class candidates ($\alpha_k = 0$, as in naive self-training) and accepting only as many as the minority class has confident candidates ($\alpha_k = 1$, i.e. full class balance, matching Section 3.8); minority-class candidates are always accepted in full. This progressively tightens the class balance of accepted pseudo-labels over the course of self-training, rather than enforcing full balance starting from round 1 as our class-balanced selection does.

We run both baselines at the same three representative imbalance levels and 8 seeds used throughout (Table 4).

4 Experiments

4.1 Setup

Both experimental tracks reuse the same $N = 1500$ -sequence, 8M-parameter-ESM-2 embedding pool and fixed 300-example test split described in Section 3.2. All self-training runs use $\tau = 0.9$ unless otherwise noted, up to 5 rounds, and the classifier and optimization settings of Section 3.

4.2 Main Multi-Label Results

Table 1 shows micro-F1 across label fractions for supervised-only, self-training, and masking-based consistency regularization on the full 10-category multi-label task.

Table 1: Micro-F1 by label fraction, full 10-category multi-label task. Self-training collapses *below* the supervised-only baseline at every fraction where it is applied; consistency regularization consistently matches or exceeds it.

Label fraction	Supervised only	Self-training	Consistency reg.
10% ($n = 120$)	0.386	0.099	0.391
25% ($n = 300$)	0.361	0.238	0.380
50% ($n = 600$)	0.368	0.292	0.395
100% ($n = 1200$)	0.381	0.381	0.381

4.3 Where Does Self-Training Collapse?

At 10% labels, the self-training classifier predicts a positive label in only 1.0% of (protein, compartment) cells on the test set, versus a true positive rate of 12.4% and 5.3% predicted by the supervised-only baseline; 270/300 test proteins receive *no* predicted compartment at all (vs. 175/300 for the baseline), which is biologically impossible since every protein has at least one true localization. Table 2 breaks this down by compartment: self-training drives F1 to exactly 0.000 on every compartment except Nucleus, the most common class, where it still collapses from 0.646 to 0.260. This observation—that even the single easiest, most balanced category is not spared—motivates asking whether collapse severity is simply monotonic in how imbalanced a category is, which we test directly in Section 4.4.

Table 2: Per-compartment F1 at 10% labels. Self-training erases nearly all signal the supervised baseline had.

Compartment	Supervised only	Self-training
Cell membrane	0.136	0.000
Cytoplasm	0.333	0.000
Endoplasmic reticulum	0.000	0.000
Extracellular	0.194	0.000
Golgi apparatus	0.000	0.000
Lysosome/Vacuole	0.000	0.000
Mitochondrion	0.000	0.000
Nucleus	0.646	0.260
Peroxisome	0.000	0.000
Plastid	0.000	0.000

4.4 Controlled Class-Imbalance Experiment: Locating the Danger Zone

An initial single-seed pilot sweeping 5 imbalance levels (5–45% positive rate) against 6 label fractions suggested that collapse was concentrated near 15% positive rate, with milder imbalance ($\geq 25\%$) appearing safe. Because this pilot used only one random seed per condition, we could not tell

whether this boundary was a real, reproducible property of the data-generating process or an artifact of one particular random draw. We therefore replicated the sweep at a fixed, previously-worst-case label fraction of 10%, at finer imbalance resolution (6–30% in 10 steps), with 5 seeds per condition, and applied the regime classification of Section 3.6 to each seed individually.

Table 3 shows the result. The reproducible danger zone is *wider* than the single-seed pilot suggested (extending up to about 20%, not stopping at 15%) but also more sharply bounded once individual seeds—rather than only condition means—are examined: the mean-based F1 gap is negative across almost the entire 6–30% range, but the fraction of individual seeds independently classified as regime B (collapse) shows a clear structure: 60–100% of seeds collapse for positive rates between 8% and 20% (peaking at 100% at 18%), while at 22% and 25% the mean gap is still negative but only 20% of seeds actually collapse (the mean is driven by a minority of runs)—indicating these should be described as occasional degradation rather than a reliable failure mode. By 30%, no seeds collapse. At the most severe imbalance we tested (6%), the baseline itself already fails to learn (regime A, mean F1 = 0.055), so self-training cannot be said to “collapse” a signal that was never there.

Table 3: Fine-grained imbalance sweep at 10% label fraction, 5 seeds per condition. *Regime* is the classification of Section 3.6 applied to the condition mean; *seed collapse fraction* is the fraction of the 5 individual seeds independently classified as regime B, which we treat as the primary evidence for how reliably collapse occurs at each level.

Positive rate	Mean F1 gap (self – base)	Regime	Seed collapse fraction
6%	−0.055	A (baseline-degenerate)	0.2
8%	−0.202	B (collapse)	0.8
10%	−0.191	B (collapse)	0.8
12%	−0.162	B (collapse)	0.6
15%	−0.277	B (collapse)	0.6
18%	−0.283	B (collapse)	1.0
20%	−0.221	D (partial degradation)	0.6
22%	−0.118	D (partial degradation)	0.2
25%	−0.138	D (partial degradation)	0.2
30%	−0.050	C (stable)	0.0

We further verified three representative levels—5% (baseline-degenerate), 15% (collapse), and 35% (stable)—with 8 seeds each (Table 4), which additionally lets us compare consistency regularization at 8-seed resolution. At 15%, self-training collapses in 6 of 8 seeds (75%), a reproducible failure rate rather than a single-run artifact; at 35%, self-training never collapses (0/8); at 5%, the baseline itself fails in the large majority of seeds. Notably, consistency regularization’s mean F1 at 15% (0.383) closely tracks the *baseline* (0.371), not the collapsed self-training result (0.072)—i.e., it remains robust precisely where self-training fails most reliably. These same three levels and seeds serve as the basis for the prior-shift control (Section 4.5) and remedy comparison (Section 4.8) below.

4.5 Ruling Out Train-Test Prior Shift as a Confound

Because the held-out test set in Section 4.4 always reflects the natural label distribution ($\approx 41\%$ positive), the manipulated training-pool imbalance and the resulting train-test distribution mismatch (prior shift) both grow as r moves away from this natural rate—confounding two possible explanations for the danger zone: is it caused by the training pool’s imbalance itself, or by the growing mismatch

Table 4: Representative-level verification, 8 seeds per condition, label fraction 10% (mean $\pm$ std). “Seed collapse fraction” is the fraction of the 8 seeds individually classified as regime B.

Positive rate	Supervised only	Self-training	Consistency reg.	Seed collapse fraction
5%	0.038 ± 0.051	0.002 ± 0.006	0.055 ± 0.053	0.12
15%	0.371 ± 0.175	0.072 ± 0.196	0.383 ± 0.188	0.75
35%	0.635 ± 0.039	0.619 ± 0.073	0.637 ± 0.036	0.00

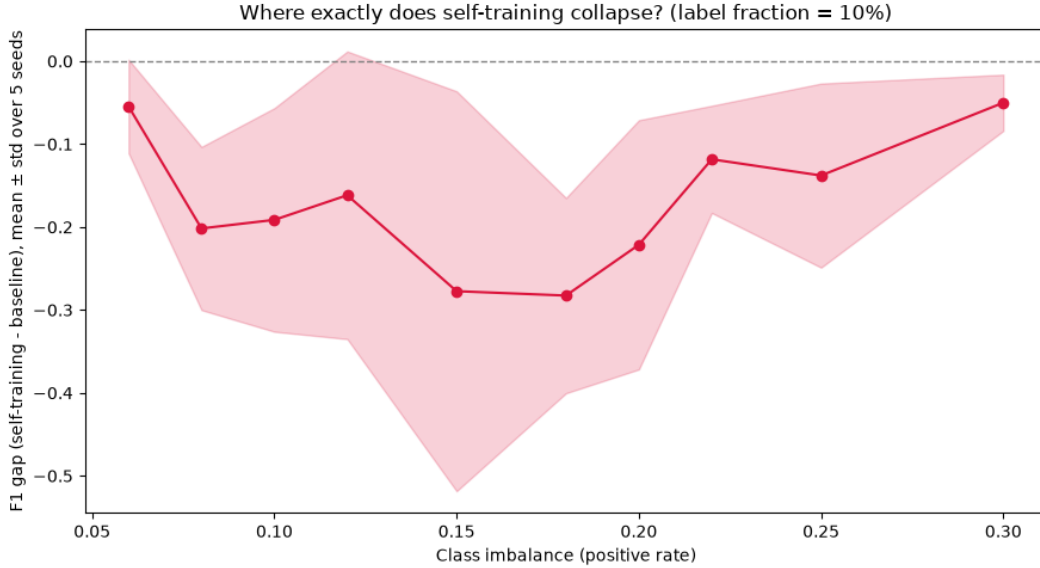


Figure 1: F1 gap (self-training – baseline) across the fine-grained imbalance sweep, mean $\pm$ std over 5 seeds. The danger zone is a bounded region, not a monotonic function of imbalance severity.

between what the classifier is trained on and what it is evaluated on? To separate them, we construct a matched-test control: at each of the three representative levels, we additionally resample a 150-example test set to the *same* target rate r as the training pool (drawn from the same fixed 300-example pool, without replacement), removing prior shift entirely, and re-run self-training with the same 8 seeds used in Table 4.

Table 5 shows the result: the collapse fraction is numerically identical between the natural-distribution and matched-distribution test sets at all three levels (12.5%, 75%, and 0%, respectively), and the F1 gaps are of comparable magnitude and sign. The danger zone is therefore a property of the training-pool imbalance itself, not an artifact of the evaluation set’s distribution.

Table 5: Matched-test-distribution control, 8 seeds per condition. “Natural” test sets reflect the fixed natural label distribution ($\approx 41\%$ positive); “matched” test sets are resampled to the same positive rate as the training pool, eliminating train-test prior shift.

Positive rate	F1 gap (natural)	Collapse fraction (natural)	F1 gap (matched)	Collapse fraction (matched)
5%	-0.036	0.125	-0.028	0.125
15%	-0.300	0.750	-0.232	0.750
35%	-0.016	0.000	-0.012	0.000

4.6 Is the Danger Zone Explained by Embedding Geometry?

A second alternative explanation is that the danger zone reflects a geometric property of the ESM-2 embedding space at that specific imbalance level—e.g., embeddings becoming more anisotropic or occupying fewer effective dimensions—rather than being purely a consequence of how imbalance corrupts pseudo-label selection during self-training. We test this using the four geometry diagnostics of Section 3.11, computed on the same fine-grained sweep and seeds as Table 3 (10 imbalance levels $\times$ 5 seeds, 50 conditions, no additional training required), applying the same partial-correlation and predictive-power tests used for the calibration causal chain (Section 4.7): for each metric, its partial correlation with the F1 gap after controlling for imbalance level, and separately whether it discriminates collapsed from non-collapsed seeds via ROC-AUC.

Table 6 shows the result. Three of the four remaining metrics—pool anisotropy, positive- and negative-class anisotropy—show essentially no relationship with the F1 gap ($|r| \leq 0.04$, $p \geq 0.77$) and do not discriminate collapsed from non-collapsed seeds (AUC ≈ 0.53). Centroid cosine similarity and effective rank each show a weak signal on one of the two tests but not the other: centroid cosine similarity’s correlation with the F1 gap is small and non-significant ($r = -0.126$, $p = 0.383$) but its predictive AUC is the highest of the five metrics (AUC = 0.659, $p = 0.082$); effective rank shows the opposite pattern, a nominally significant partial correlation ($r = -0.319$, $p = 0.024$) but a non-significant predictive AUC (AUC = 0.650, $p = 0.091$). We do not treat either as a confirmed finding: with five metrics tested simultaneously, none of the ten reported p -values (five correlation, five AUC) survives a Bonferroni correction ($\alpha \approx 0.01$); the two tests disagree with each other for both metrics showing any signal; and the effective-rank values themselves vary only narrowly across imbalance levels (98.3–99.3, against an embedding dimensionality of 384), suggesting a small effect size even if the association were real. We report this conservatively as weak, mutually inconsistent signals warranting further investigation rather than a second causal mechanism. Together with the matched-test-distribution control (Section 4.5), this indicates the danger zone is a property of imbalance-driven pseudo-labeling dynamics rather than of the embedding space itself.

Table 6: Embedding geometry diagnostics vs. the F1 gap, computed on the fine-grained sweep (10 imbalance levels $\times$ 5 seeds, 50 runs). Partial r controls for imbalance level via residualization; AUC tests whether the metric discriminates collapsed from non-collapsed seeds. None of the five metrics survives a Bonferroni correction for testing five metrics simultaneously ($\alpha \approx 0.01$).

Metric	Naive r	Partial r	p (partial)	AUC (collapse)	p (AUC)
Pool anisotropy	−0.036	−0.039	0.789	0.534	0.582
Positive-class anisotropy	−0.043	−0.024	0.866	0.531	0.655
Negative-class anisotropy	−0.031	−0.032	0.827	0.526	0.583
Centroid cosine similarity	−0.126	−0.126	0.383	0.659	0.082
Effective rank	−0.315	−0.319	0.024	0.650	0.091

4.7 Calibration Precedes Collapse: A Quantified Causal Chain

Figure 2 shows round-by-round ECE and test F1 for the three representative levels (5 seeds, mean $\pm$ std). At 15%, ECE rises sharply within the first 2–3 self-training rounds while F1 falls over the same rounds; at 35%, both remain flat and stable across all 5 rounds; at 5%, ECE is already high and F1 already near zero from round 1 (the baseline-degenerate floor effect).

Beyond this visual trend, we test the causal chain quantitatively at the seed level using the ECE-escalation criterion of Section 3.6. Across all 50 runs of the fine-grained sweep (Table 3’s conditions $\times$ 5 seeds), the Pearson correlation between ECE escalation (last-round ECE minus round-1 ECE) and the F1 gap is $r = -0.91$ (Figure 3)—a strong, robust relationship holding across the full range of tested conditions, not just at isolated points.

At the seed level (Table 7), we find $P(\text{ECE escalates} \mid \text{F1 collapses}) = 100\%$ at both levels where collapse occurred (5%: 1/1; 15%: 6/6): in our controlled experiment, *every* instance of self-training collapse was accompanied by significant calibration escalation. The converse is not true, however: at 15%, ECE escalates in 100% of seeds (8/8) but F1 collapses in only 75% (6/8), and at 35%, ECE escalates in a quarter of seeds (2/8) with *no* accompanying F1 collapse. We conclude that ECE escalation is a *necessary but not sufficient* precursor to collapse in this setting, and phrase our causal claim accordingly rather than treating the relationship as symmetric.

This correlation could in principle be driven entirely by imbalance level itself acting as a common cause of both ECE escalation and the F1 gap, rather than reflecting a direct relationship between the two. We test this by computing a partial correlation: residualizing both ECE escalation and the F1 gap on imbalance level (via linear regression) before correlating the residuals. The partial correlation is $r = -0.92$ ($p < 10^{-20}$), effectively unchanged from—if anything, marginally stronger than—the naive $r = -0.91$: the calibration-collapse relationship is not an artifact of imbalance level as a confound. Separately, we test whether ECE measured before self-training begins (round 1, prior to any pseudo-labeling) predicts which seeds will later collapse: it does not (ROC-AUC = 0.55, effectively chance; two-sample $t = 0.20$, $p = 0.84$). Together, these results sharpen our causal claim: the calibration-collapse relationship concerns how ECE evolves *during* self-training, not a static property of the initial, pre-training classifier.

Table 7: Seed-level joint statistics for ECE escalation and F1 collapse, 8 seeds per condition, label fraction 10%. ECE escalation is defined as an absolute rise ≥ 0.05 from round 1 to the final self-training round.

Positive rate	ECE escalation rate	F1 collapse rate	Both co-occur	$P(\text{ECE} \uparrow \mid \text{F1 collapse})$
5%	62%	12%	12%	100%
15%	100%	75%	75%	100%
35%	25%	0%	0%	n/a (no collapse events)

4.8 Comparing Remedies: Calibration, Threshold Tuning, and Class-Balanced Pseudo-Label Selection

A calibration-based explanation for collapse predicts a specific remedy: correcting confidence before thresholding should prevent it. We test this directly against two alternative explanations—that collapse is simply a poorly chosen threshold, or that it reflects imbalanced pseudo-label *selection* independent of calibration—by running calibrated self-training (Section 3.7), the τ -ablation, and class-balanced selection (Section 3.8) across the same three representative levels and 8 seeds (5 seeds for the τ -ablation) used in Table 4.

Table 8 shows calibrated self-training and class-balanced selection across all three levels. Calibrated self-training meaningfully reduces the collapse fraction at 15% relative to naive self-training (Table 4: 75%) but does not eliminate it (37.5%), while the fitted temperature ($T \approx 2.24$ at 15%, and consistently $T > 1$ at all three levels) confirms the classifier is genuinely overconfident, validating the calibration diagnosis itself. Class-balanced selection, in contrast, eliminates collapse entirely at

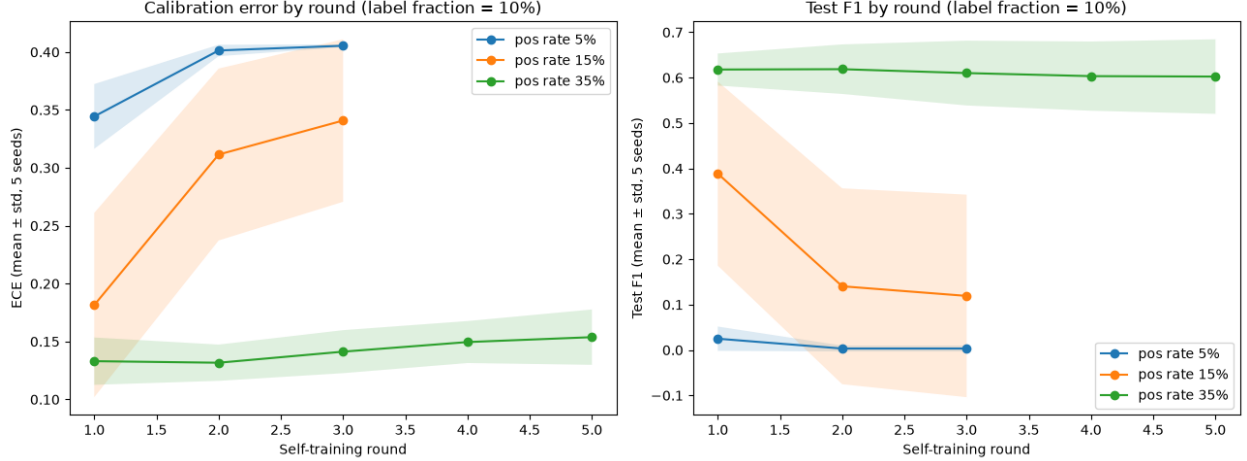


Figure 2: Round-by-round ECE (left) and test F1 (right) at three representative imbalance levels, mean $\pm$ std over 5 seeds. Only the collapse-regime level (15%) shows both a sharp ECE rise and a simultaneous F1 drop within the first few rounds.

every level tested (0% collapse fraction) and slightly *exceeds* the supervised baseline at 15% and 35%.

Table 8: Calibrated self-training and class-balanced pseudo-label selection across the three representative levels, 8 seeds per condition (mean values). T is the mean fitted temperature at the final self-training round; $T > 1$ indicates the uncalibrated classifier is overconfident.

Positive rate	Supervised only	Calibrated self-tr.	Mean T	Class-balanced self-tr.	Collapse fraction (cal. / bal.)
5%	0.038	0.000	1.86	0.038	0.125 / 0.000
15%	0.371	0.202	2.24	0.426	0.375 / 0.000
35%	0.635	0.611	2.93	0.674	0.000 / 0.000

Table 9 isolates the effect of threshold choice alone (no calibration, no balancing) at the danger-zone level (15%, a separate 5-seed replication). Every tested τ still shows substantial collapse (20–60% of seeds); the most conservative threshold ($\tau = 0.99$) reduces collapse to 20% but does not eliminate it, and does so by accepting very few pseudo-labels per round.

Table 9: Confidence-threshold ablation at the danger-zone level (15% positive rate, label fraction 10%), 5 seeds per condition, no calibration. No tested threshold eliminates collapse.

τ	Self-training F1	F1 gap	Collapse fraction
0.70	0.246	−0.142	0.4
0.80	0.150	−0.238	0.6
0.90	0.111	−0.277	0.6
0.95	0.139	−0.250	0.6
0.99	0.281	−0.107	0.2

Table 10 summarizes all four candidate explanations side by side at the danger-zone level. Class-balanced selection is the only remedy that fully eliminates collapse; calibration and threshold tuning both provide partial, incomplete relief. This ranking indicates that while miscalibration is real

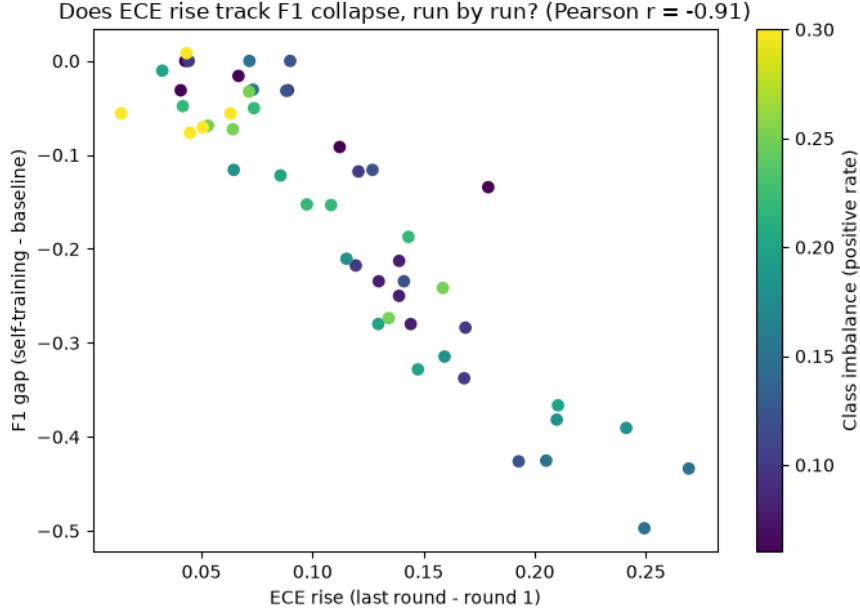


Figure 3: ECE escalation vs. F1 gap across all 50 runs of the fine-grained sweep (Pearson $r = -0.91$; partial $r = -0.92$ after controlling for imbalance level), colored by imbalance level.

and calibration helps, the more direct lever on collapse is *which* pseudo-labels get accepted across classes, not confidence calibration per se—consistent with, and providing independent evidence for, the class-rebalancing insight in the general imbalanced-SSL literature [Kim et al., 2020, Wei et al., 2021] (Section 2), now demonstrated as a direct causal fix in this setting rather than only a correlational one.

Table 10: Synthesis: four candidate explanations/remedies at the danger-zone level (15% positive rate). Class-balanced selection is the only one that fully eliminates collapse.

Method	F1 gap	Collapse fraction
Uncalibrated self-training ($\tau = 0.9$)	-0.300	75%
Calibrated self-training	-0.169	38%
Best threshold alone ($\tau = 0.99$)	-0.107	20%
Class-balanced selection ($\tau = 0.9$)	+0.055	0%

4.9 Comparison Against Established Class-Imbalanced SSL Baselines: DARP-style and CReST-style

The remedies compared above (Table 10) are either our own diagnostic ablations or directly inspired by, but not identical to, methods from the general class-imbalanced SSL literature (Section 2). We now test the literature methods themselves: DARP-style distribution alignment and CReST-style progressive rebalancing (Section 3.12), run at the same three representative levels and 8 seeds as Table 4.

Table 11 extends the danger-zone (15%) synthesis to six methods. The result is unexpected: despite being given an oracle advantage—the true target class ratio, which none of the other methods

are told and which would be unknown in a real deployment—the DARP-style baseline collapses *more* often than even naive, uncalibrated self-training (88% vs. 75% of seeds), and has the most negative mean F1 gap of all six methods tested (-0.365). A plausible mechanism is that DARP-style alignment only corrects the *aggregate* marginal probability over all currently-unlabeled examples to match the target rate; it does not improve the ranking of individual examples by correctness likelihood. The scalar bias needed to shift the aggregate marginal toward the (low) target positive rate can push previously sub-threshold, genuinely uncertain minority-class predictions over the acceptance threshold, injecting confidently-wrong pseudo-labels into the self-reinforcing loop at a higher rate than an unshifted threshold would—the aggregate distribution becomes correct while individual acceptance decisions do not improve, which is precisely what our class-balanced selection targets instead by re-ranking candidates *within* each class by confidence rather than shifting a global threshold. The CReST-style curriculum performs closer to, but does not improve on, naive self-training (75% collapse, gap -0.293 vs. -0.300): because its curriculum starts at full naive acceptance ($\alpha_1 = 0$) and only tightens gradually, and Section 4.7 shows miscalibration escalates within the first few rounds, most of the damage plausibly accrues before the curriculum’s balancing constraint becomes strict enough to matter.

We emphasize that this comparison should not be read as showing our method is superior to DARP or CReST in general: both were designed for multi-class, long-tailed classification, and our binary adaptations (Section 3.12), particularly DARP’s oracle target-rate assumption, are simplifications that may disadvantage them relative to their original, more sophisticated multi-class formulations. What the comparison does show is that, in this controlled binary setting, aligning an aggregate distribution or applying a slowly-tightening curriculum is not sufficient to fix collapse, while directly controlling the per-round, per-class quality of accepted pseudo-labels is—reinforcing, with a negative result for two literature-inspired alternatives, that pseudo-label selection quality rather than global distributional statistics is the more direct lever in this failure mode.

Table 11: Six-way comparison at the danger-zone level (15% positive rate, 8 seeds per condition). DARP-style is given an oracle advantage (the true target class ratio) that the other methods do not have, yet performs worst of all six.

Method	F1 gap	Collapse fraction
Uncalibrated self-training ($\tau = 0.9$)	-0.300	75%
Calibrated self-training	-0.169	38%
Best threshold alone ($\tau = 0.99$)	-0.107	20%
Class-balanced selection (ours)	$+0.055$	0%
DARP-style (oracle target rate)	-0.365	88%
CReST-style (progressive curriculum)	-0.293	75%

4.10 Reproducibility Across PLM Scale

All experiments above use a single 8M-parameter ESM-2 checkpoint. To test whether the danger zone, and the mechanism behind it, are specific to this small checkpoint, we repeat the fine-grained imbalance sweep, the three-level 8-seed representative verification, and the ECE-F1 causal-chain analysis using a four-times-larger checkpoint, `facebook/esm2_t12.35M_UR50D` (35M parameters), reusing the identical protein sample and train/pool/test split (Section 3.10).

Table 12 summarizes the comparison. The danger zone reproduces at nearly the same location and shape at 35M as at 8M: the fine-grained sweep shows the same regime pattern (strong collapse

concentrated in roughly 8–18% positive rate, a partial-degradation gray zone around 20–25%, and stability from 30% on). At the representative levels, the 5% and 35% conditions are essentially unchanged between scales (12% collapse fraction at 5% for both; 0% at 35% for both), while at the danger-zone level (15%) the collapse fraction is attenuated but not eliminated at the larger scale (75%→62% of seeds). The calibration-collapse causal chain (Section 4.7) is nearly identical across scales: the naive correlation between ECE escalation and F1 gap is $r = -0.91$ at 8M vs. $r = -0.874$ at 35M, the partial correlation controlling for imbalance level is $r = -0.92$ at 8M vs. $r = -0.890$ at 35M, and $P(\text{ECE escalates} \mid \text{F1 collapses}) = 100\%$ at both scales.

Table 12: 8M vs. 35M ESM-2 checkpoint comparison, identical protein sample and data split. Collapse fraction is reported at the three representative levels (8 seeds each); correlations use the full fine-grained sweep (10 levels $\times$ 5 seeds) at each scale.

Quantity	8M (<code>esm2_t6</code>)	35M (<code>esm2_t12</code>)
Collapse fraction at 5%	12%	12%
Collapse fraction at 15% (danger zone)	75%	62%
Collapse fraction at 35%	0%	0%
ECE escalation vs. F1 gap, naive r	-0.91	-0.874
ECE escalation vs. F1 gap, partial r	-0.92	-0.890
$P(\text{ECE} \uparrow \mid \text{F1 collapse})$	100%	100%

We conclude that the danger zone and its calibration-mediated mechanism reproduce across at least this 8M-to-35M scale range, with collapse severity moderating somewhat as scale increases but not disappearing—evidence against the possibility that our central finding is an artifact of using an unusually small, low-capacity PLM checkpoint, while leaving open whether the effect continues to shrink, plateaus, or vanishes at substantially larger scale (hundreds of millions of parameters and beyond), which we leave to future work (Section 5).

4.11 Consistency Regularization Remains Robust in the Danger Zone

Across every condition in Tables 1, 3, and 4, masking-based consistency regularization matches or exceeds the supervised-only baseline, including at the 15% positive-rate level where self-training collapses in 75% of seeds (Table 4: consistency reg. 0.383 ± 0.188 vs. baseline 0.371 ± 0.175 vs. collapsed self-training 0.072 ± 0.196). Because this method never thresholds soft predictions into hard pseudo-labels, it has no confidence threshold for miscalibration to corrupt—consistent with our diagnosis that miscalibrated hard pseudo-labeling, specifically, drives collapse. Notably, consistency regularization and class-balanced selection (Section 4.8) are the only two methods in this study that fully avoid collapse in the danger zone, despite addressing the problem through entirely different mechanisms: never thresholding into hard labels at all, versus thresholding but balancing which labels are accepted.

4.12 Random-Guess Reference

For context, a random classifier that predicts each compartment positive at its true prevalence rate achieves micro-F1 ≈ 0.124 by construction; a coin-flip (50%) random classifier achieves micro-F1 ≈ 0.199 . The supervised-only baseline (0.386) is well above chance, while collapsed self-training (0.099) falls *below* even the prevalence-matched random baseline.

5 Discussion and Limitations

Single target category. Our controlled imbalance experiment (Section 4.4) binarizes a single compartment (Nucleus) to keep the manipulation and its interpretation tractable; the full multi-label setting (Table 1) confounds ten simultaneously varying, correlated imbalance levels. Repeating the controlled protocol for other compartments, and eventually for the joint multi-label imbalance structure, is a natural and important extension; our matched-test-distribution control (Section 4.5) and remedy comparison (Section 4.8) were likewise only tested for this single category.

Only two, both comparatively small, PLM checkpoints. Section 4.10 shows the danger zone and its calibration-mediated mechanism reproduce, with mild attenuation, between an 8M- and a 35M-parameter ESM-2 checkpoint. This rules out the phenomenon being an artifact of our original, smallest checkpoint, but both checkpoints are still small relative to the largest publicly available ESM-2 models (up to 15B parameters) and to other PLM families entirely (e.g., ProtBERT, ProtT5). Whether collapse severity continues to shrink, plateaus, or eventually vanishes at substantially larger scale remains an open question; the mild but incomplete attenuation we observe across a four-fold parameter increase is at least suggestive that scale alone is unlikely to eliminate the failure mode within a practical range.

Binary simplification of DARP and CReST, and DARP’s oracle assumption. The DARP-style and CReST-style baselines of Section 4.9 are our own binary-classification adaptations of methods originally designed for multi-class, long-tailed classification (Section 3.12); their relative underperformance here should not be read as a general claim about DARP or CReST, which may adapt differently or perform better under a more faithful multi-class implementation, additional hyperparameter tuning, or a different SSL backbone. The DARP-style baseline in particular is given an oracle target class ratio that is unknown in realistic deployments and would need to be estimated, which if anything should have advantaged it relative to the other methods compared; that it still collapses more often than naive self-training is a substantive negative result for this specific adaptation, not evidence that distribution-aligning refinement is unhelpful in its originally intended, multi-class setting.

Weak, inconsistent geometric signals not further investigated. Section 4.6 finds that four of five embedding-geometry diagnostics show no relationship with collapse, while effective rank and centroid cosine similarity each show a weak signal on one of two statistical tests but not the other, neither surviving correction for testing five metrics simultaneously. We treat this as an honest null result for the geometric-explanation hypothesis rather than as evidence that these two metrics are entirely uninformative; a dedicated, adequately powered follow-up (e.g., a larger grid of imbalance levels and seeds, or a pre-registered single-metric test) would be needed to confirm or rule out a genuine but small effect.

Calibration is a genuine but partial mechanism. Section 4.8 shows that calibrated self-training reduces but does not eliminate collapse, while class-balanced pseudo-label selection eliminates it entirely. This indicates that miscalibration, while real (fitted temperatures are consistently $T > 1$) and causally implicated (Section 4.7), is not the sole lever on collapse; biased pseudo-label *selection* under imbalance appears to be the more direct cause, with miscalibration acting as a contributing but incomplete mediator. We did not test combining calibration with class-balanced selection, which could plausibly close the remaining gap; we leave this to future work.

Necessary, not sufficient, and dynamically emerging. As established in Section 4.7, ECE escalation reliably accompanies collapse but does not reliably predict it in the opposite direction (ECE escalated in 100% of seeds at 15% positive rate, while only 75% actually collapsed), and this relationship survives controlling for imbalance level as a potential confound (partial $r = -0.92$ vs. naive $r = -0.91$). However, ECE measured before self-training begins does not predict later collapse

(AUC= 0.55)—the diagnostic signal lies in how calibration evolves *during* self-training, not in the initial model. Practitioners should not treat a well-calibrated starting point as reassurance, nor rising ECE alone as a certain early-warning signal; what distinguishes escalated-but-non-collapsed seeds from escalated-and-collapsed ones remains an open question.

Calibrated self-training’s labeled-data cost. Withholding 25% of the labeled budget for calibration (Section 3.7) means calibrated self-training trains its classifier on less data than the other compared methods at the same nominal label fraction—a real-world cost of calibration that any deployment would need to weigh, and a possible confound in the comparison of Table 8 that a more careful ablation (e.g., matching effective training-set size across methods) should address.

Fixed, pre-specified thresholds. The regime-classification and remedy thresholds used throughout ($F1 < 0.10$ for baseline-degeneracy, $0.5\times$ for collapse, $[0.8, 1.2]\times$ for stability, 0.05 absolute ECE rise for escalation, the 25% calibration-holdout fraction, and the confidence-threshold grid of Section 4.8) were chosen once, in advance, rather than tuned to produce a particular narrative, but they remain design choices rather than derived quantities. We report them explicitly so they can be scrutinized and varied; a sensitivity analysis over these thresholds is a natural robustness check for future work.

Single-seed pilots can mislead. Our own initial single-seed sweep (which motivated Section 4.4) placed the danger zone narrowly around 15% and suggested 25%+ was uniformly safe; the 5-seed replication revised this to a wider 8–20% zone with 22–25% as an occasional-degradation gray zone rather than either “dangerous” or “safe.” We flag this as a methodological point in its own right: characterizing an SSL failure mode from a single run, even when the qualitative direction is correct, can meaningfully mischaracterize its boundary.

Macro-F1 on rare compartments. Macro-F1 on rare compartments (e.g., Peroxisome, Plastid) is inherently unstable at our sample size ($N = 1500$) and should be interpreted cautiously in the multi-label track.

6 Conclusion

We set out to characterize, rather than merely observe, when and why self-training collapses on frozen protein language model embeddings. Using a controlled experiment that directly manipulates class imbalance while holding the embedding space, architecture, and evaluation set fixed, and applying seed-level statistical rigor rather than relying on single runs, we find that collapse is concentrated in a specific, identifiable, non-monotonic imbalance regime (roughly 8–20% positive rate in our setup)—not simply more likely the more imbalanced a class is—and confirm with a matched-test-distribution control that this is a property of training-pool imbalance itself, not an artifact of train-test distribution mismatch. We show that this collapse is reliably preceded by rising miscalibration: every collapse event co-occurred with significant ECE escalation, this relationship holds even after controlling for imbalance level (partial $r = -0.92$), and escalation is a dynamically emerging signal rather than one visible from the initial, pre-training model (round-1 ECE does not predict later collapse). Directly comparing three candidate remedies, we find that calibration helps but does not fully resolve the failure, and that explicitly balancing which pseudo-labels are accepted across classes is the most effective intervention tested, eliminating collapse entirely—indicating that biased pseudo-label selection under imbalance, not miscalibration alone, is the more direct mechanistic lever, and connecting our diagnosis to the general class-rebalancing insight of the imbalanced-SSL literature. We show that a simple, PLM-native remedy—masking-based consistency regularization, which never converts soft predictions into thresholded hard pseudo-labels—remains robust throughout this danger zone without any explicit calibration correction. We then test the

robustness of these findings and two alternative explanations. Repeating the fine-grained sweep and causal-chain analysis with a four-times-larger ESM-2 checkpoint under an identical data split reproduces the danger zone at nearly the same location with the calibration-collapse relationship intact, and only mild attenuation in collapse severity, indicating the phenomenon is not an artifact of our smallest checkpoint. Four embedding-geometry diagnostics show no relationship with collapse severity that survives controlling for imbalance level and correcting for multiple comparisons, ruling out embedding geometry as an independent explanation. Finally, directly comparing class-balanced selection against binary-adapted DARP-style and CReST-style baselines from the class-imbalanced SSL literature, we find both underperform it—DARP-style even more so than naive self-training, despite an oracle advantage unavailable in practice—indicating that per-round, per-class control over pseudo-label quality, rather than aggregate distribution alignment or a progressive curriculum, is the more direct lever on this failure mode. While our experiments use protein subcellular localization as a testbed, the underlying question—under what conditions does naive self-training fail catastrophically, what causally drives that failure, and which interventions actually fix it—is a general one for semi-supervised learning, and we hope our controlled-manipulation protocol, quantitative failure-regime criteria (Section 3.6), and remedy comparisons prove useful beyond this domain.

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